

From Coarse-Grained to Atomic Resolution: Multiscale Simulations Reveal How TDP-43 Condensates Are Regulated

Biomolecular condensates play central roles in cellular organization, yet how they form and are regulated within crowded, cell-scale environments remains poorly understood. Here we show how the protein Hero11 regulates TDP-43 biomolecular condensates, using large-scale coarse-grained (CG) simulations that reveal condensate organization and Hero11's modulatory effects. Representative structures are then back-mapped to all-atom resolution for extended MD, yielding atomic-level mechanistic insights inaccessible to CG models alone. This multiscale pipeline is built on CG models integrated into GENESIS's MD engines, achieving high performance on the Fugaku supercomputer, illustrating how HPC-driven multiscale simulation can tackle disease-relevant biomolecular assemblies.